

Hackathon Problem Statement

Multimodal Data Management Pipeline for RAG-Ready Systems

Duration: 13 hours **Theme:** Building structured, RAG-ready data from multimodal sources

1. The Problem

Modern organizations generate huge amounts of information through videos, audio recordings, images, PDFs, presentations, screen recordings, and documents. While this data contains valuable knowledge, most of it is not directly usable by modern RAG systems.

Current RAG pipelines are largely designed around text. A document is split into chunks, embedded, stored, and retrieved. This becomes problematic when knowledge is distributed across multiple modalities.

For example, in a recorded technical meeting, a speaker may explain an architecture while a diagram is displayed on screen. Important details may exist in the speaker's words, the diagram, the text inside the diagram, and the exact point in the recording where the discussion occurred. Treating each of these independently causes important relationships and context to disappear.

The challenge is therefore not simply to extract text from multimedia. The challenge is to build a data management pipeline capable of turning raw multimodal information into structured, meaningful, and retrieval-ready knowledge while preserving the relationships between different pieces of evidence.

The system should be able to answer questions such as: *"What architecture was discussed for reducing database load, who explained it, and where was the corresponding diagram shown?"* Answering this correctly may require connecting information from speech, video frames, OCR, timestamps, and related documents.

2. Core Scope (Mandatory)

A. Multimodal Ingestion

The input should contain realistic information rather than isolated toy examples. The system should extract useful information from each modality while retaining its relationship to the original source.

- Video
- Audio
- Images
- PDFs or documents

B. Structured Knowledge Representation

Extracted information must be represented in a form that can be consumed by downstream retrieval systems. A transcript sentence, an image, a video frame, and a piece of OCR text should not necessarily be treated as unrelated chunks when they represent the same underlying information.

- Content
- Modality
- Timestamp
- Source
- Entities
- Relationships

- Confidence
- Provenance

C. Cross-Modal and Temporal Relationships

The system must preserve relationships between information appearing across modalities. For example, when a speaker discusses an architecture while an architecture diagram is visible, the system should be able to associate the spoken explanation with the corresponding visual evidence.

- Relationships between speech and visual evidence
- Temporal location of information
- Connections between related observations across modalities

D. Retrieval

Queries should be capable of retrieving relevant evidence from multiple modalities rather than relying exclusively on transcript or text similarity. Retrieved information should remain connected to its original source.

- Search or query over the resulting multimodal corpus
- Retrieval across multiple modalities
- Traceability to original sources

3. Explicitly Out of Scope

- Training custom foundation models
- Building ASR, OCR, or vision models from scratch
- Production-grade distributed infrastructure
- Building a vector database from scratch
- Full authentication and authorization
- Supporting every possible media format
- Highly polished frontend experiences

Existing models, APIs, databases, and open-source libraries may be used. The focus is the data-management and retrieval problem.

4. Stretch Goals

- Automatically identifying semantic events instead of relying on fixed-size chunks.
- Building relationships between entities appearing across different files and modalities.
- Providing exact provenance for retrieved evidence down to timestamps or image regions.
- Retrieving a combination of text, images, frames, and other evidence for a single query.
- Building a temporal knowledge graph over the processed data.
- Comparing the system against conventional text-only RAG and demonstrating measurable retrieval improvements.
- Supporting questions about how information changes over time.
- Introducing confidence scores for extracted information and using them during retrieval.

5. Suggested Technical Approach

Teams are free to choose their architecture and technology stack. A possible implementation could use existing media-processing tools for video and audio extraction, ASR for speech, OCR and vision models for images, and embedding models for semantic representation.

The resulting information can be stored using any combination of relational databases, vector databases, graph databases, or search indexes.

The important design question is not which technology is used, but how the system represents and connects information so that downstream retrieval can recover the correct context.

6. Evaluation Criteria

Criterion	Weight
Multimodal ingestion quality	20%
Quality of the resulting data representation	20%
Cross-modal and temporal context preservation	20%
Retrieval quality	20%
Technical depth and innovation	15%
Demo quality	5%

Judges will evaluate whether the system actually solves the underlying problem rather than simply converting multimedia into text and performing ordinary vector search.

A strong submission should demonstrate a query where the required evidence exists across multiple modalities and show that the system can retrieve and connect that evidence correctly.

7. Deliverables

- Working repository with README
- Working multimodal ingestion pipeline
- Structured representation of processed data
- Query or retrieval interface
- Demonstration using at least three modalities
- Evaluation against a basic text-centric RAG approach
- Architecture explanation and key design decisions
- Brief description of future improvements

8. Timeline Suggestion (13 Hours)

Hr 0–1	Understand the problem, choose datasets and architecture.
Hr 1–5	Build multimodal ingestion and extraction.
Hr 5–8	Build the structured representation and cross-modal relationships.
Hr 8–11	Implement retrieval and connect it to the processed data.

Hr 11–12	Testing, evaluation, and integration.
Hr 12–13	Demo preparation and final polish.

The final demo should show a realistic multimodal dataset being ingested and a query whose answer requires information from more than one modality.